

MIDTERM I

ECO403/503–INTRODUCTION TO CONTRACT THEORY AND AUCTIONS

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Instructions. The exam has a total of 100 points. The points of each question are shown in boldface. The exam starts at 1:30 PM and finishes at 2:45 PM, i.e., you have 75 minutes. ***You need to submit your written solutions in pdf format to Moodle before 2:50 PM. Solutions sent by email will not be accepted, so plan accordingly to submit on time.***

1. **(40 points)** Consider a decision maker (DM) that has preference over lotteries, with $Z \subseteq \mathbb{R}_{++}$ being the set of monetary prizes¹. Let $v : Z \rightarrow \mathbb{R}$; then the DM assigns the number $v(z)$ to each amount of money $z \in Z$. The function v is continuous and increasing, and strictly positive. Define $U : \mathcal{L}(Z) \rightarrow \mathbb{R}$ by

$$U(p) = \alpha \max\{v(z) \mid p(z) > 0\} + (1 - \alpha) \min\{v(z) \mid p(z) > 0\},$$

where $0 \leq \alpha \leq 1$. Let $\succsim$ denote the preference relation on $\mathcal{L}(Z)$ that is represented by U .

- (a) **(20 points)** Does the DM's preference over lotteries $\succsim$ admit a Expected Utility representation?
- (b) **(20 points)** The notion of *risk aversion* is independent of Expected Utility Theory, so irrespective of your answer to the previous item, we can define aversion as follows.² A DM is risk averse if she always prefers the expected value of any lottery with certainty to the lottery itself, that is, for any $p \in \mathcal{L}(Z)$, $[E(p)] \succsim p$. Characterize decision makers with this type of preference that are risk averse for any α and v .
2. **(20 points)** A price-taking firm, i.e., operating under perfect competition, seeks to maximize profits. The firm produces a good y , which sells it at price p , and faces costs of production given by $c(y, w_1, w_2) = y^2(w_1 + w_2)$, where w_1 and w_2 are the prices of two inputs in the production of y . However, the firm faces uncertainty regarding the price p : with probability λ the price is p_h and with probability $1 - \lambda$ the price is p_l , with $p_h > p_l$.
- (a) **(10 points)** State the firm's problem and solve for the supply function of the firm, i.e., the optimal value of y as a function of the prices and λ . Be as rigorous as you can.
- (b) **(10 points)** Now suppose that p is a continuous random variable that can take values on the interval $[p_l, p_h]$ and is distributed according to the probability density function f . State the firm's problem and solve for the supply function. Be as rigorous as you can.
3. **(40 points)** Consider a situation where all individuals face an idiosyncratic risk³ to lose 100 with probability p . Suppose there are N individuals. The individuals decide to create a mutual agreement, where the aggregate loss in the pool will be equally split among its members.
- (a) **(10 points)** Describe explicitly in terms of lotteries the residual loss⁴ of each member of the pool when $N = 2$ and $N = 3$.
- (b) **(10 points)** Give a formal proof that an increase from $N = 2$ to $N = 3$ generates a reduction in risk borne by each member in the sense of Second Order Stochastic Dominance.
- (c) **(20 points)** Generalize the result for the case of N and $N + 1$, where N is an arbitrary but finite number of individuals.

¹The notation $\mathbb{R}_{++}$ denotes the strictly positive part of the real line.

²Notation: $E(p)$ denotes the expected value of the lottery p and $[y]$ denotes the degenerate lottery that awards y with probability one.

³Idiosyncratic: adjective denoting peculiar or individual. Thus, idiosyncratic risk means that each individual faces this risk and that the probability of one individual facing this risk is independent of the others.

⁴If an individual with wealth w losses an amount L , the residual loss is $-L$.